



COLLEGE OF ENGINEERING



Fuels



Characteristics of a Good Fuel

Suitability

- The fuel selected should be most suitable for the process. Eg. coke made out of bituminous coal is most suitable for the blast furnace.

High Calorific Value

- The fuel should possess high calorific value because calorific value determines the efficiency of the fuel.

Ignition temperature

- Ignition temperature is the minimum temperature to which the fuel must be heated so that it starts burning smoothly. A good fuel should have a moderate ignition temperature. If the fuel has a low ignition temperature, it can lead to fire hazard. If the fuel has high ignition temperature, then a lot of heat will be required to heat it but its storage and transportation would become easier.

Characteristics of a Good Fuel

Moisture Content

- The moisture content should be low because presence of high moisture lowers the effective calorific value of the fuel.

Non combustible matter content

- The non-combustible matter forms clinkers after combustion. The presence of the non combustible matter reduces the heating value of the fuel.

Nature of the products

- The products formed should not be harmful. They should not cause unpleasant odours and other harmful effects.

Characteristics of a Good Fuel

Cost of fuel

- The fuel should have a low cost and should be available in large amount at a cheap rate.

Smoke

- Formation of the smoke should be as low as possible which is must for a healthy environment.

Control of the process

- The fuel should be such that its combustion can be easily controlled.

Solid Fuels

Solid Fuels which occur naturally are **primary fuels**. For example: wood and different types of coal constitute the important primary solid fuels.

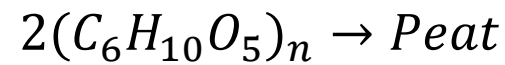
The solid fuels which are derived from the natural varieties are **secondary fuels**. For example charcoal and coke.

Let us discuss some of the important solid fuels like wood and coal.

- Wood has been used as a fuel from ancient times.
- However, due to deforestation, wood is no longer used except in forest areas where wood is available at a very low cost.
- The calorific value of air-dried wood is about 3,500 to 4,500 kcal/kg.
- It burns with a long and non smoky flame, leaving a small amount of ash. Although ash content is low, the oxygen content is very high. This makes , even dry wood a fuel of low calorific value.
- Wood is largely used as a domestic fuel owing to its low ignition temperature, non-smoky flame and absence of ash disposal problems.

Coal is regarded as a fossil fuel produced from the vegetable debris under conditions of high temperature and pressure over a million of years.

- The transformation of the vegetable debris to coal takes place in two stages:
 - ❖ **Biochemical or peat stage:** In this step, the plant materials are attacked by various microorganisms. The end product of this stage is peat.



- ❖ **Chemical Stage or metamorphism:** In this stage, the peat deposit buried under sedimentary deposits lose moisture and volatile components under the effect of high temperature and pressure. The peat gets enriched in carbon whereas its oxygen content decreases. The spongy peat transforms into hard brittle coal gradually.

Classification

Coals are mainly classified on the basis of their rank, which is defined as the degree of coalification from wood.

Type of Coal	Percentage (dry, mineral matter free basis)					% moisture	Calorific value (cal/g)
	C	H	O	N	VM		
Wood	40-50	5-6	20-40	0-0.5	--	70-90	4000-4500
Peat	45-60	3.5-6.5	20-45	0.75-3	45-75	70-90	4125-5280
Brown Coal	60-75	4.5-5.5	17-35	0.75-2	45-60	30-50	6600-7100
Bituminous Coal	5-90	4.0-5.5	20-30	0.75-2	11-50	10-20	6600-8800
Anthracite	90-95	3-4	2-3	0.5-2	3.8-10	1.5-3.5	8470-8800

Analysis of Coal

The composition of coal varies according to the degree of coalification. Coal can be analysed in two ways:

- **Proximate analysis**
- **Ultimate analysis**
- **Proximate Analysis:** It is an empirical but important analysis. The data varies with the procedure adopted and hence it is called proximate analysis. It gives information about the practical utility of coal. In this analysis we determine moisture, ash, volatile matter and fixed carbon.

Proximate Analysis

- 1) **Moisture Content**: The original coal is always associated with moisture. When a wet coal is exposed to the atmosphere, the external moisture evaporates but the apparently dry coal still contains some moisture. This moisture is referred to as air dried moisture. Air dried moisture is determined by heating a known amount of coal to 105 to 110°C in an electric oven for about an hour. After one hour, it is taken out from the oven and cooled in the desiccator and weighed.

Loss in weight of coal is reported as moisture content on percentage basis.

Percentage of moisture = Loss in weight/weight of coal taken x 100



Significance

- Excess of moisture is undesirable in coal.
- Moisture lowers the heating value of coal and the heat is wasted in the form of latent heat of vaporisation. For every percent of moisture present, 1% of heat is lost.
- Moisture also increases the transport costs.
- Excessive surface moisture may cause difficulty in handling the coal.

Hence, lower the moisture content, better is the quality of coal as a fuel.

Proximate Analysis

2) **Volatile Matter:** The volatile matter comprises of complex mixture of gaseous and liquid products resulting from the thermal decomposition of the coal. It consists mainly of combustible gases such as H_2 , CO , CH_4 and other hydrocarbons.

It is determined by heating a known weight of moisture free coal sample in a covered platinum crucible at $950^\circ C$ for 7 minutes. The crucible is cooled, first in air, then inside a desiccator and weighed. Loss in weight is reported as volatile matter on percentage basis.

% of Volatile Matter = Loss of weight due to removal of volatile matter / Weight of coal sample taken x 100

- A high percentage of volatile matter in fuel indicates that a large portion of the fuel is burnt as gas or vapour or even may escape unburnt.
- The high volatile matter content gives long flames, high smoke.
- Coals containing less than 14% of volatile matter do not cake at all and are thus not suitable for manufacturing coke.

However, high volatile matter content is desirable in coal gas manufacture because volatile matter in a coal denotes the proportion of the coal which will be converted into gas and tar products by heat.

Proximate Analysis

3) Ash: Coal contains inorganic mineral substances which are converted into ash by chemical reactions during the combustion of coal. Ash usually consists of silica, alumina, iron oxide and small quantities of lime, magnesia, etc.

Ash content is determined by heating the residue left after the removal of volatile matter at approximately 700°C for half an hour without covering. The cooled mass is weighed. From the weight of the residue, the % of ash is calculated.

$$\% \text{ of ash} = \frac{\text{Weight of the residue left}}{\text{Weight of the coal}} \times 100$$

Significance

- The high percentage of ash is undesirable.
- Ash may restrict the passage of air and lower the rate of combustion.
- High ash leads to large heat losses and cause problems of clinkering (forming of ash lumps). It has been estimated that one percent rise in coal ash is equivalent to 0.3 to 4% decrease in boiler efficiency.
- Apart from loss of efficiency of coal, clinker formation also leads to loss of fuel because some coal particles also get embedded in the clinkers.

Proximate Analysis

4) Fixed Carbon: Fixed carbon content increases from low ranking coals such as lignite to high ranking coals such as anthracite. Higher the percentage of fixed carbon greater is its calorific value and better is the quality of coal. This represents the quantity of carbon that can be burnt by a primary current of air.

The percentage of fixed carbon is given by:

% of fixed carbon = 100 - [% of moisture + volatile matter + ash]

Significance

- Higher the percentage of fixed carbon, greater is the calorific value.
- The percentage of fixed carbon helps in designing the furnace and shape of the fire-box because it is the fixed carbon that burns in the solid state.

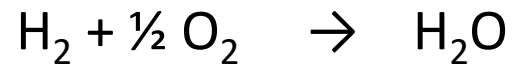
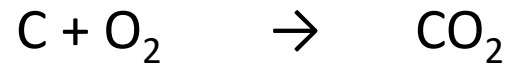
Numerical Based on the Analysis

Q1. A sample of coal was analysed as follows: Exactly 2.000g was weighed into a silica crucible. After heating for one hour at 110°C, the residue weighed 1.975g. The crucible next was covered with a vented lid and strongly heated for exactly seven minutes at 950°C. The residue weighed 1.328 g. The crucible was then heated without the cover, until a constant weight was obtained. The last residue was found to weigh 0.205 g. Calculate the percentage results of the above analysis.

Ultimate Analysis

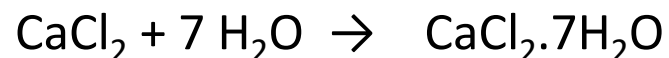
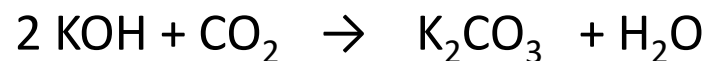
Ultimate Analysis includes the estimation of carbon, hydrogen, sulphur, nitrogen and oxygen.

1. Carbon and Hydrogen: A known amount of coal is taken in a combustion tube and is burnt in excess of pure oxygen. The carbon and hydrogen present in the fuel are converted to CO_2 and H_2O respectively.



The gaseous products of combustion are passed through two bulbs, one containing weighed amount of anhydrous CaCl_2 which absorbs water, and the other containing weighed amount of KOH which absorbs carbon dioxide. The percentage of carbon and hydrogen are calculated from the increase in weight of the respective absorption tubes.

Calculations



Let, weight of coal sample taken = x g

Increase in weight of CaCl_2 tube = z g

Increase in weight of KOH tube = y g

44g of CO_2 contain = 12g of carbon

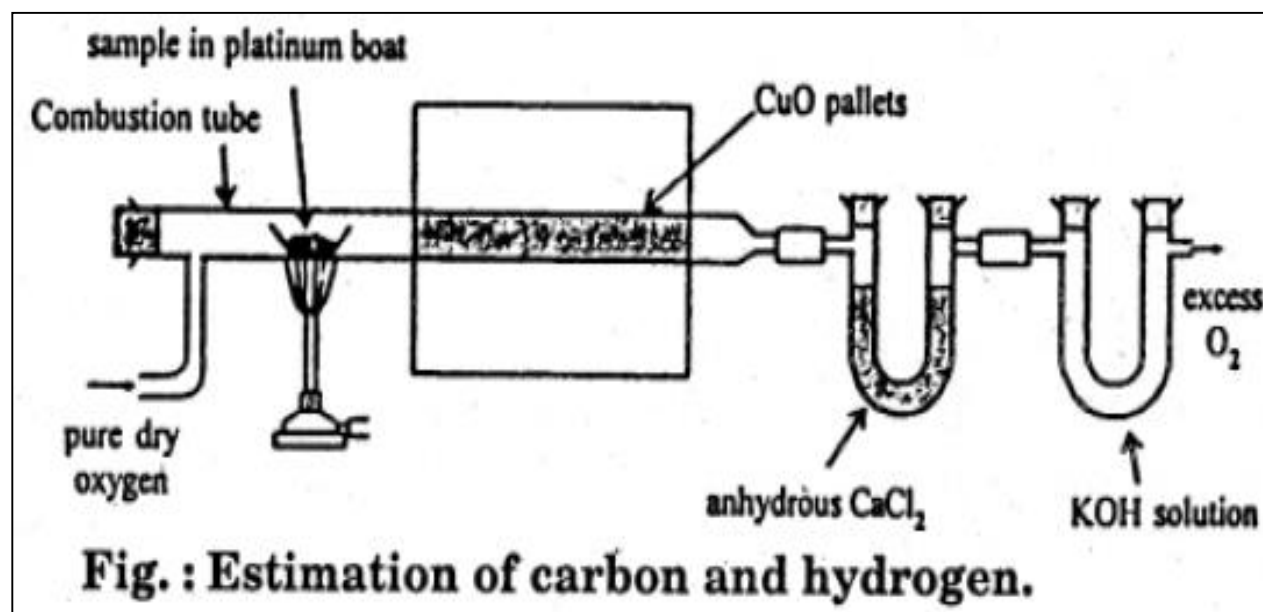
Y g of CO_2 contain = $12/44 \times y$ g of carbon

% of carbon = $(12/44 \times y)/\text{weight of coal taken} \times 100$

18 g of water contain = 2 g of hydrogen

z g of water contain = $2/18 \times z$ g of hydrogen

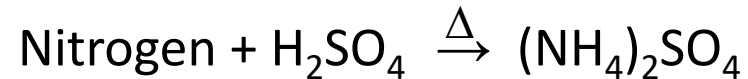
% of hydrogen = $2/18 \times z/\text{weight of coal taken} \times 100$



Significance

- Calorific value of a fuel is directly related to its carbon content. Hence greater the percentage of carbon, greater is the calorific value of the fuel.
- A higher percentage of carbon also reduces the size of the combustion chamber.
- Higher percentage of hydrogen also increases the calorific value of coal. The hydrogen of coal is responsible for the production of many useful materials, eg., gaseous hydrogen, methane and coal chemicals in carbonization and gastification in industries.

2. Nitrogen : Nitrogen present in the coal sample can be estimated by Kjeldahl's method. A known weight of the given coal sample along with $K_2SO_4 + HgSO_4$ which acts as a catalyst is heated with conc. H_2SO_4 . Nitrogen present in coal gets converted to $(NH_4)_2SO_4$.



- The contents are then transferred to a round bottom flask and the solution is heated with excess of NaOH.
- The NH_3 gas thus liberated is absorbed in a known volume of a standard solution of acid.
- The unused acid is then determined by titrating with NaOH. From the volume of acid used by NH_3 liberated, the percentage of nitrogen can be calculated.

Calculations



Let the weight of coal taken = W g

Volume of the acid used up = V_1 mL

Normality of the acid = N_1

$$V_1 \text{ mL of } N_1 \text{ acid} = V_2 \text{ mL of } N_2 \text{NH}_3$$

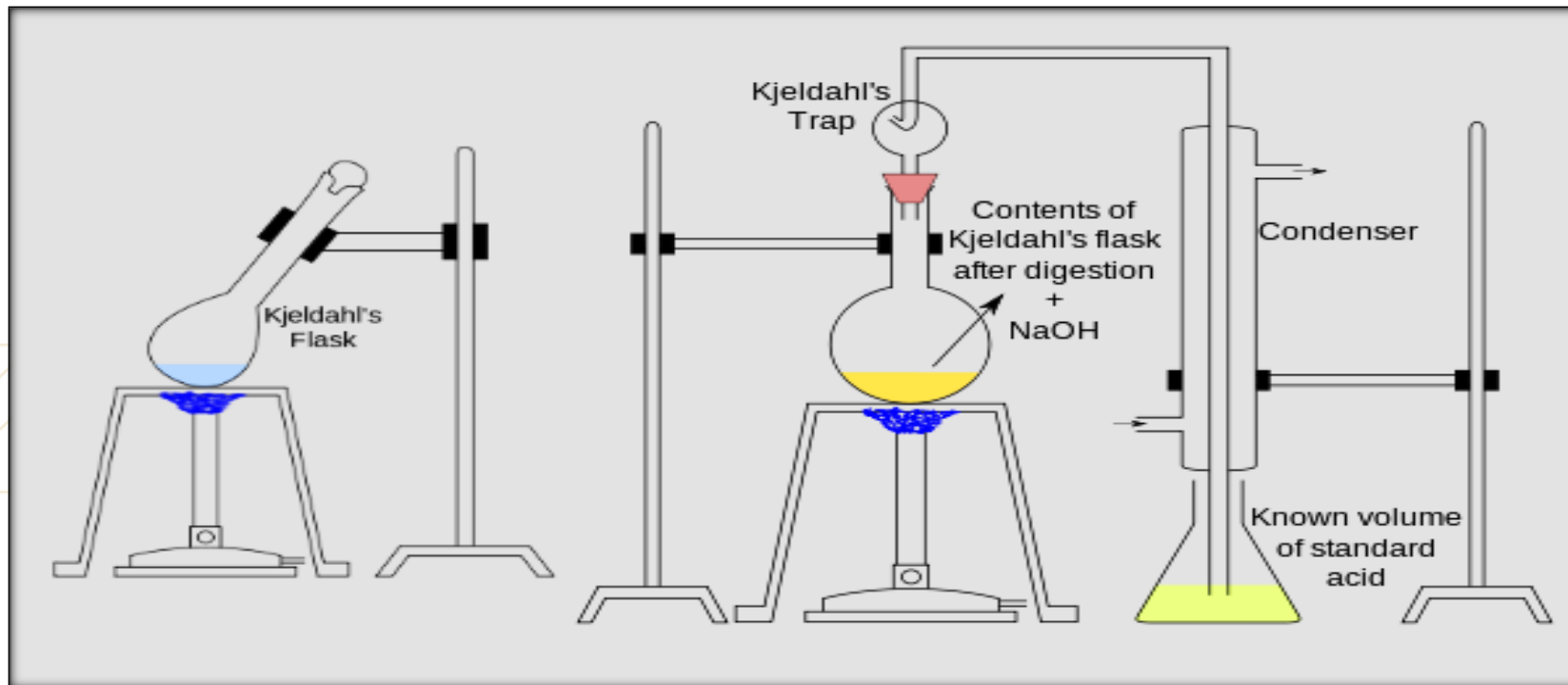
$$V_2 \text{ mL of } 1\text{N NH}_3 = V_1 \times N_1 \text{ mL of acid}$$

But 1000 ml of 1N NH_3 solution = 17g of NH_3 = 14g of Nitrogen

Calculations

V_2 mL of 1 N NH_3 solution = $[14/1000] \times N_1 \times V_1$ g of Nitrogen
(since $V_2 = N_1 V_1$ ml)

Percentage of nitrogen = $[14/1000] \times [N_1 \times V_1 \times 1000 / W]$
 $= 1.4 \times N_1 \times V_1 / W$



Significance

- Nitrogen content in most of coals is between 1 and 2 percent. It has no calorific value and hence its presence is undesirable.
- However, it is useful in carbonization industries which recover a part of it as valuable by products like NH_3 , $(\text{NH}_4)_2\text{SO}_4$ and pyridine bases.

3. Sulphur: Sulphur present in the coal is oxidised by fuming nitric acid into sulphuric acid.

- The sulphuric acid is precipitated as BaSO_4 by the addition of BaCl_2 , quantitatively.
- The precipitate of BaSO_4 so obtained from the above method is filtered and heated to a constant weight. From the weight of BaSO_4 obtained, the sulphur present in the coal is calculated.

Let the weight of coal sample = W g

The mass of the precipitate = W_1 g

Ultimate Analysis

Since 233 g of $\text{BaSO}_4 = 32$ g of sulphur

W_1 g of $\text{BaSO}_4 = 32/233 W_1$ g

% of sulphur = $[32/233] \times [W_1/W] \times 100$

Significance: Sulphur is an undesirable constituent as it emits corrosive SO_2 , SO_3 during combustion and highly objectionable H_2S during carbonization of coal. These gases cause atmospheric pollution. Sulphur containing coal is not suitable for the preparation of metallurgical coke as it adversely affects the properties of metal.

Ultimate Analysis

4. **Oxygen**: The oxygen is usually determined by the method of difference. The carbon, hydrogen, nitrogen and sulphur are expressed in percent basis and their sum is subtracted from 100.

$$\% \text{ of Oxygen} = 100 - (\% \text{ of carbon} + \text{hydrogen} + \text{Nitrogen} + \text{sulphur})$$

Significance: Lesser the oxygen content, better is the coal. This is because oxygen content of coal is associated with moisture, low calorific value and low coking power (discussed later)

Numerical Based on the Analysis

Q1. 1.0 g of a sample of coal was used in a bomb calorimeter for the determination of calorific value. Calorific value of coal was found to be 8,800cal/g. The ash formed in the bomb calorimeter was extracted with acid and the acid extract was heated with barium nitrate solution and a precipitate of barium sulphate was obtained. The precipitate was filtered, dried and weighed. The weight of the precipitate was found to be 0.08 g. Calculate the percentage of sulphur in the coal sample.

Numerical Based on the Analysis

Q2. 1.56g of a sample of coal was kjeldhlized and NH_3 gas thus evolved was absorbed in 50 ml of 0.1N H_2SO_4 . After absorption, the excess (residual) acid required 6.25 ml of 0.1N NaOH for exact neutralization. 2.60g of the coal sample in a quantitative analysis gave 0.1755g of BaSO_4 . Calculate the percentage of N and S in the coal sample.

THANK YOU

